

Lecture 4 Live Problems

Vector fields and flow. Trajectories in state space.

Points toward lab manual pages: **ST6, ST7, ST8** (section L1 S3)

1. Isle Royale. In 1980 there are 664 moose and 50 wolves. In 1981 there are 650 moose and 30 wolves. Let M be the number of moose and W the number of wolves. Describe the change during the time interval from 1980 to 1981.

(a) What is ΔM ? _____

(b) What is ΔW ? _____

(c) What is Δt ? _____

(d) What is the *relative* change in the numbers of moose and wolves?

2. Modeling predators and prey. Our basic assumptions about moose and wolves on Isle Royale:

- More moose = more wolf food, good for wolves!
- More wolves = more moose eaten, bad for moose!
- Moose would happily flourish, eating grass and stuff.
- Wolves need to eat some big prey. Otherwise they will die out.

Make a diagram to illustrate the influences of the quantities M and W on themselves and each other. Use this to create a reasonable system of change equations.